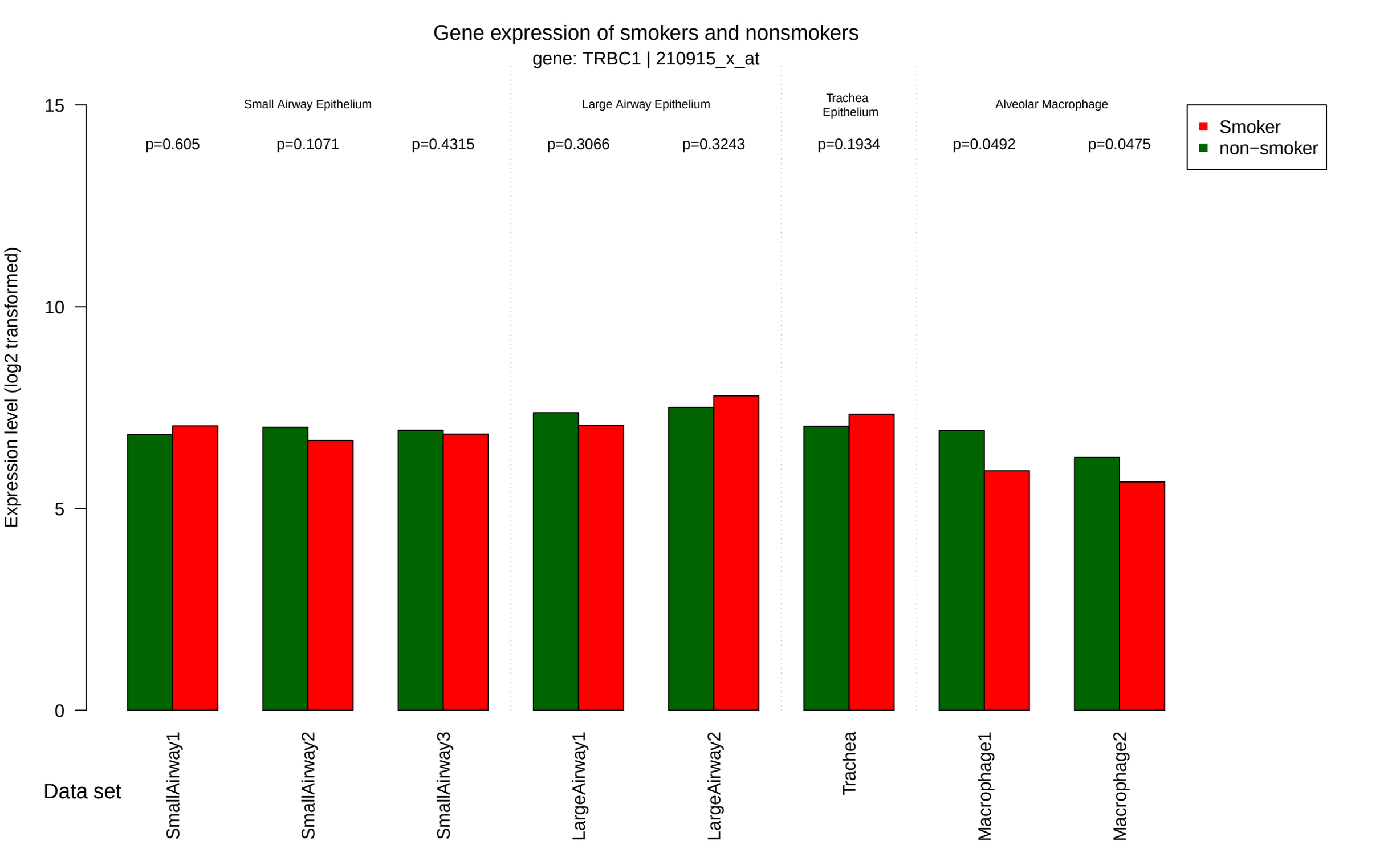
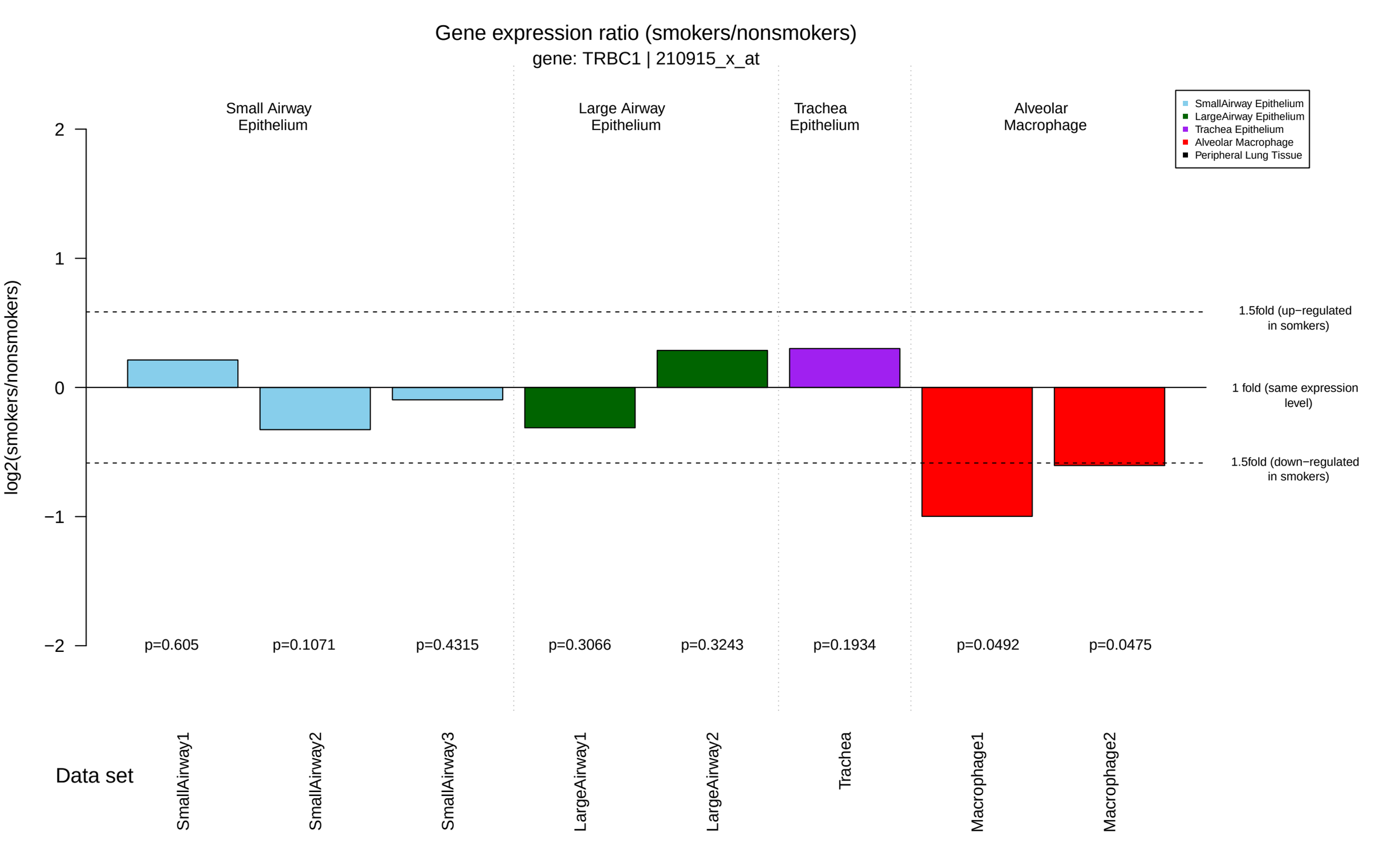
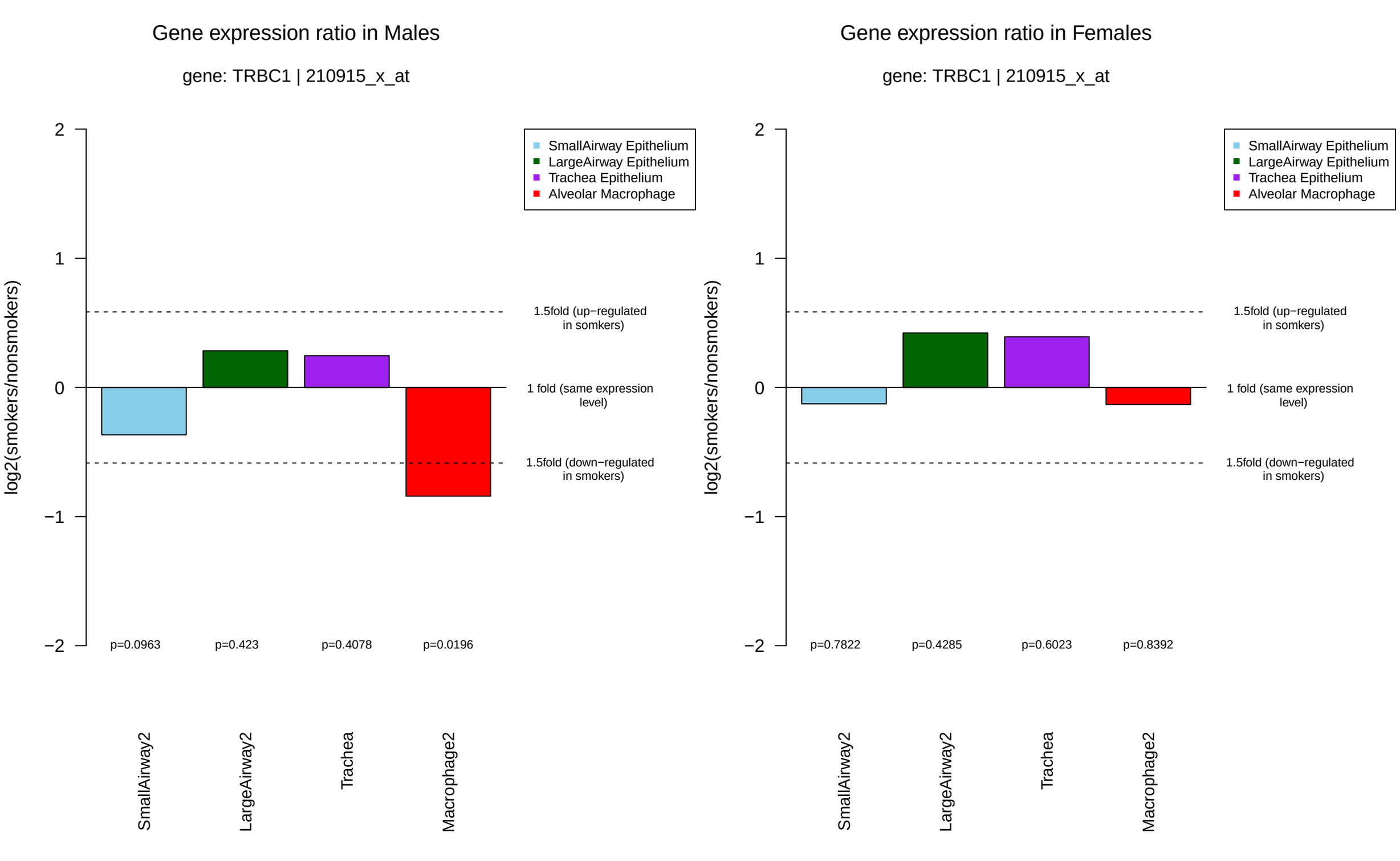
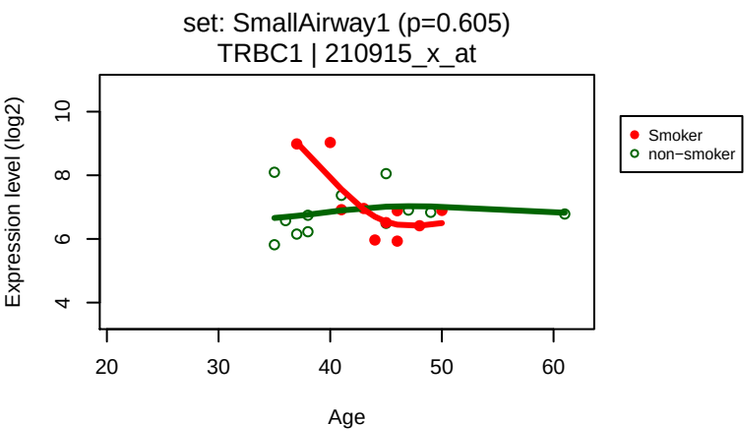
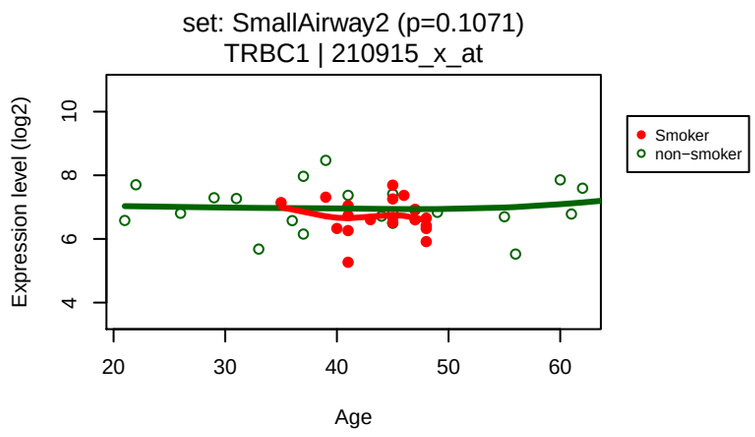
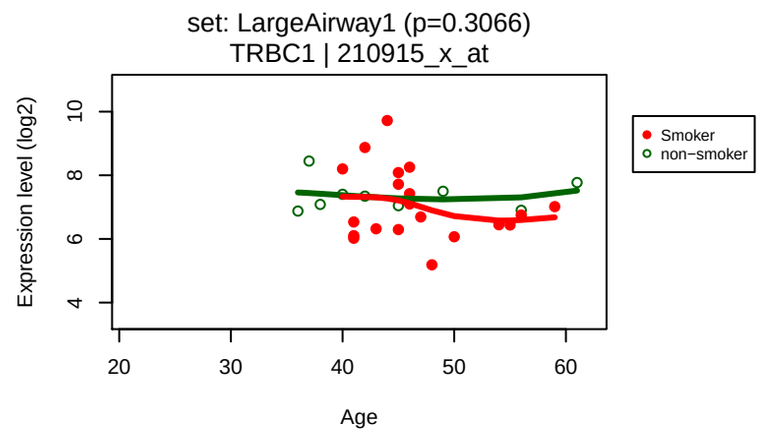

Gene expression of smokers and nonsmokers
gene: TRBC1 | 210915_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.605
p=0.1071
p=0.4315
p=0.3066
p=0.3243
p=0.1934
p=0.0492
p=0.0475
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

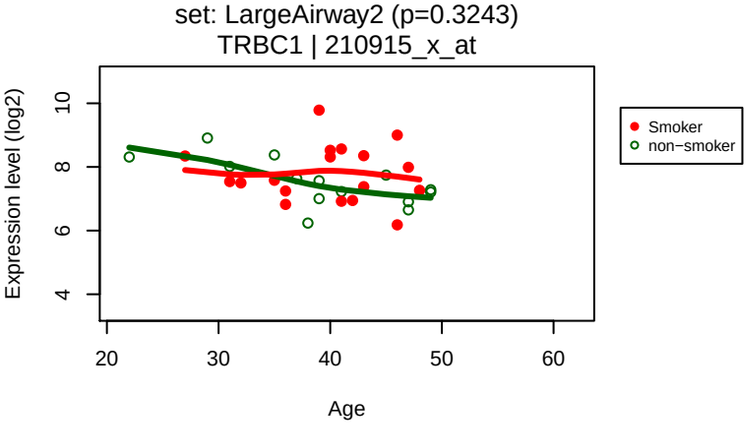

Gene expression ratio (smokers/nonsmokers)
gene: TRBC1 | 210915_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.605
p=0.1071
p=0.4315
p=0.3066
p=0.3243
p=0.1934
p=0.0492
p=0.0475
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

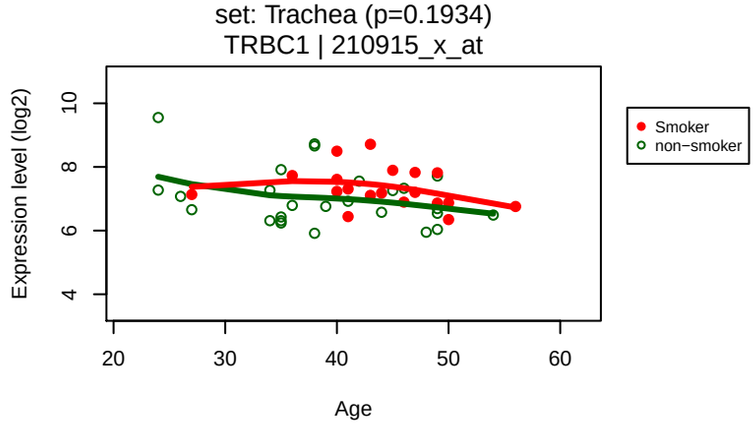

Gene expression ratio in Males
gene: TRBC1 | 210915_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0963
p=0.423
p=0.4078
p=0.0196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRBC1 | 210915_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7822
p=0.4285
p=0.6023
p=0.8392
SmallAirway2
LargeAirway2
Trachea
Macrophage2

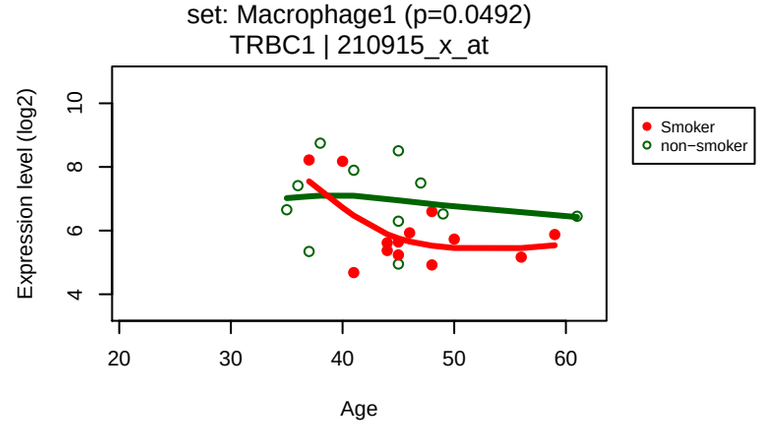

set: SmallAirway1 (p=0.605)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker

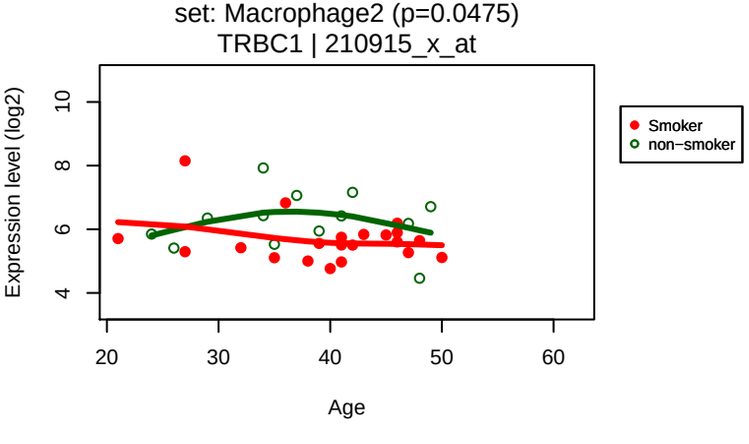

set: SmallAirway2 (p=0.1071)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker

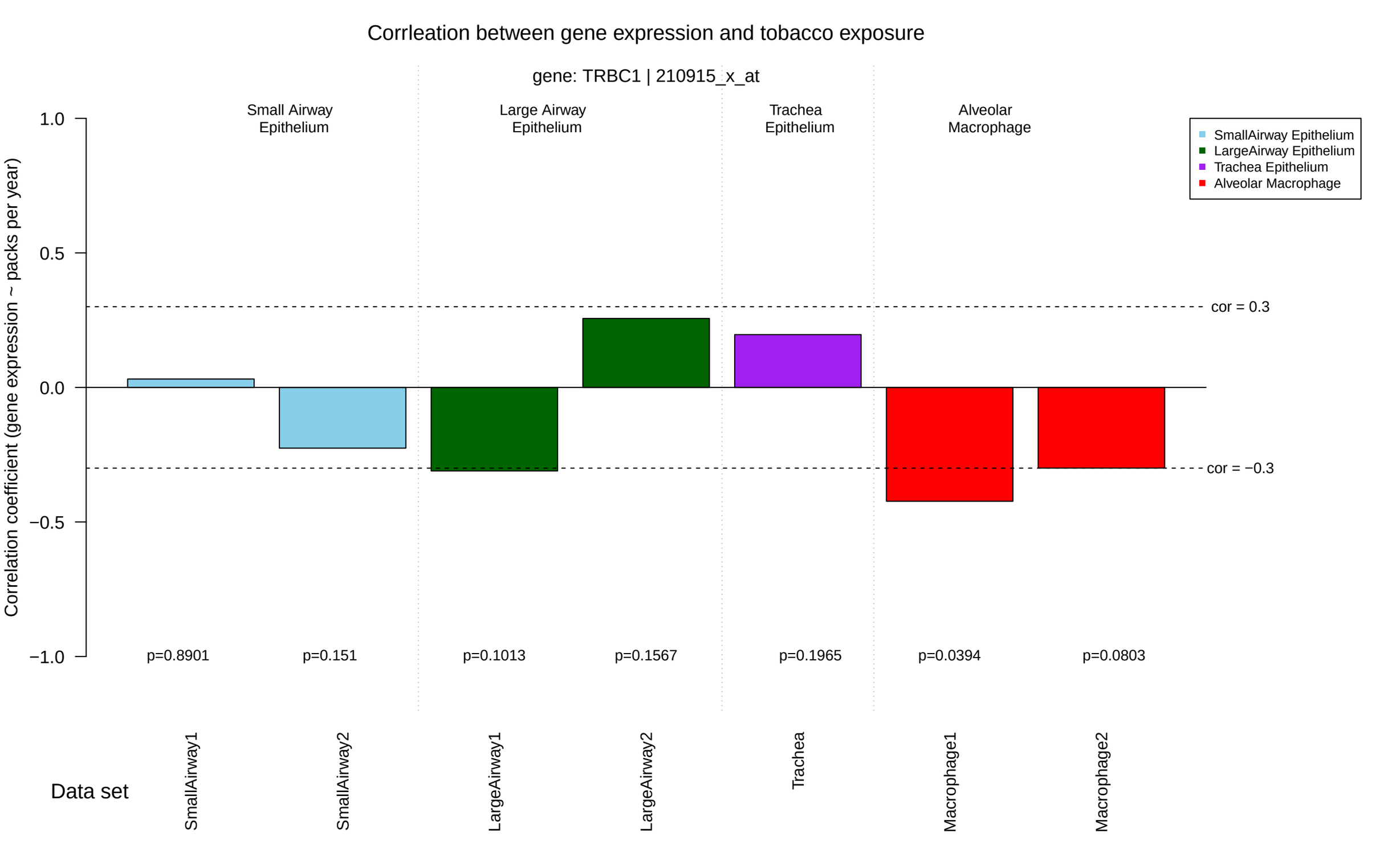

set: LargeAirway1 (p=0.3066)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3243)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1934)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0492)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0475)
TRBC1 | 210915_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRBC1 | 210915_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8901
p=0.151
p=0.1013
p=0.1567
p=0.1965
p=0.0394
p=0.0803
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2